

Tour Length Comparison - 100-City Maps

Only tour length is shown. Source 100-200 values are sorted within each panel by BlindBlendACO tour length: 1-101.

Algorithm colors

BaselineACO

BlindBlendACO

ElitistACO

MaxMinACO

Tour length averages

With Cross Check

BaselineACO: 16,010

BlindBlendACO: 15,623

ElitistACO: 15,941

MaxMinACO: 15,710

Without Cross Check

BaselineACO: 16,364

BlindBlendACO: 15,826

ElitistACO: 16,182

MaxMinACO: 15,973

With Cross Check

This line chart displays the tour length for four algorithms across 101 sorted maps/runs. The y-axis, labeled 'Tour length', ranges from 14,332 to 18,618. The x-axis, labeled 'Sorted map / run index', ranges from 1 to 101. Four data series are plotted: BaselineACO (blue), BlindBlendACO (purple), ElitistACO (red), and MaxMinACO (green). Each series includes a solid line with markers and a corresponding dashed horizontal line representing the average. BaselineACO shows the highest variability and ends with the highest value (~17,760). BlindBlendACO shows the lowest variability and ends with the lowest value (~16,903). ElitistACO and MaxMinACO show intermediate performance with significant fluctuations.

Without Cross Check

This line chart displays the tour length for the same four algorithms across 101 sorted maps/runs, but without a cross-check. The axes and data series are identical to the 'With Cross Check' chart. In this scenario, the overall tour lengths are generally higher, particularly for the later indices. BaselineACO reaches the highest value (~18,618) at the end of the run. BlindBlendACO remains the lowest, ending at ~16,903. The fluctuations in the other two algorithms are also more pronounced compared to the 'With Cross Check' scenario.

Sorted map / run index